Autonomous Systems and Robotics 700 Assignment

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Student ITS No: _____

Qualification: _____ Year of Study: _____ Semester: _____

Assignment due date: _____ Date submitted: _____

QUESTION	EXAMINER MARKS	MODERATOR MARKS	REMARKS

ASSIGNMENT INSTRUCTIONS

Please tick each box to confirm completion.

Use Times New Roman font, size 12, with 1.5 line spacing throughout the document. Apply Harvard Referencing Style for all citations and references.

For essay-style assignments, please include the following sections:

Table of Contents

Introduction

Main Body (with relevant subheadings)

Conclusion

References

Submit the assignment in PDF format on Moodle.

Use the specified cover page provided.

Include a signed declaration of originality.

DECLARATION OF ORIGINALITY:

I hereby declare that this assignment is my own work and has not been copied from any other source except where due acknowledgment is made. I affirm that all sources used have been properly cited and that this submission complies with the institution's policies on academic integrity and plagiarism.

Student Signature: **Date:**

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Contents and introduction

Contents — Introduction and objectives (p. 2); Component A methodology (p. 3); Component A results: perception and planning (p. 4); control and navigation (p. 5); Component B methodology (p. 6); swarm and learning results (p. 7); ethics and responsible autonomy (p. 8); conclusions and references (p. 9).

Problem and objective. An autonomous robot must perceive unknown obstacles, find a feasible route and control movement; a team of autonomous agents must cooperate over an imperfect communication network. This project supplies one reproducible Python reference implementation plus two Webots world/controller configurations corresponding to Components A and B (Richfield Graduate Institute of Technology, n.d.a).

Conceptual basis. The study guide defines perception as transforming raw sensor streams into actionable spatial representations (Unit 2), introduces weighted-graph search with Dijkstra and A* (Unit 3), and addresses PID control and robot kinematics (Unit 4). Units 5 and 6 address consensus, swarm intelligence, distributed systems and neural decision-making (Richfield Graduate Institute of Technology, n.d.b).

Research design. The offline reference runs with seeded synthetic measurements and outputs CSV/JSON plus Matplotlib figures. Webots worlds and controllers are provided for follow-on validation. No Webots launch or hardware trial was performed in this execution environment. Accordingly, offline figures are not presented as Webots outcomes. Experiment configurations and the exact commands are in README.txt.

Evaluation criteria. Component A compares path length, expanded graph nodes, execution time, PID dynamics and tracking error. Component B considers consensus spread, PSO coverage cost, classification accuracy and centralised/distributed coverage communication trade-offs. Risks, limitations and South African deployment considerations are analysed independently.

Component A: methodology (A1–A4)

A1: Sensing and late fusion. A 6 × 6 m occupancy map has 0.10 m cells, wall segments and red landmarks. Offline LiDAR generates 180 range measurements, with a 3.5 m maximum and assumed 0.012 m Gaussian distance noise. Log-odds mapping updates free ray cells (−0.30) and endpoint occupancy (+0.85). In camera frames, a 5 × 5 Gaussian blur, dual-range red HSV thresholding, morphological opening and 60-pixel contour-area filter produce landmark detections. Camera pixel bearing is matched to the nearest five LiDAR beams and their minimum finite range is used to add labelled map evidence. These choices combine camera semantics with metric LiDAR depth, as explained in Unit 2 (Richfield Graduate Institute of Technology, n.d.b).

A2: Graph search. The world is discretised into an occupancy grid; obstacles are inflated by 0.18 m to approximate robot clearance. Dijkstra (zero heuristic) and A* (Euclidean heuristic) share the same priority-queue implementation, eight-neighbour movement and corner-cutting prohibition. With diagonal moves, Euclidean distance is admissible. Search length is the sum of cardinal/diagonal edge costs times cell size. Nodes expanded and median of seven Python timings are measured on identical start/goal cells.

A3: Kinematic PID. The ideal differential drive obeys $dx/dt = v\cos\theta$, $dy/dt = v\sin\theta$ and $d\theta/dt = \omega$. Wheel rates are $(v \ \omega L/2)/r$, with $L = 0.160$ m, $r = 0.033$ m. Angular error is wrapped to $[-\pi, \pi]$. The PID controller clips output to ±2 rad/s and freezes its integral during saturation (anti-windup). Its step experiment assumes 0.35 s actuator lag and +0.15 rad/s disturbance; these are illustrative modelling assumptions, not fitted Webots motor parameters.

A4: Integration. The reference path follower tracks a known inflated map at 20 Hz and reports driven distance, goal error and cross-track distance. The separate Webots controller is designed to acquire camera and LiDAR at every fifth 32-ms simulation step, update its occupancy map, replan on observed obstruction, and issue differential wheel speeds each step. GPS/IMU are treated as simulation-ground-truth pose; this is not a deployed localisation solution.

Component A: perception and planning

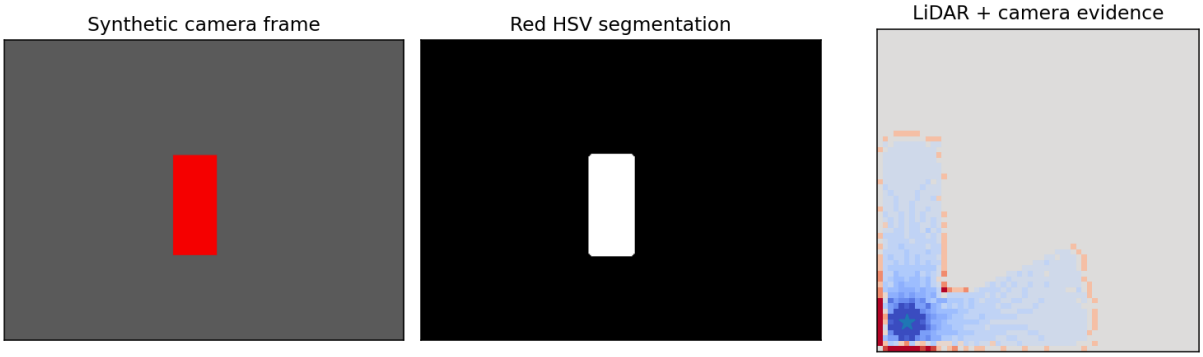


Figure 1. Synthetic camera image, red segmentation and one fused LiDAR/camera landmark. This verifies the offline processing chain, not real Webots camera calibration.

The synthetic frame produced 1 detected red region and 1 projected landmark. HSV segmentation can fail in poor illumination, shadows or on non-red obstacles. Camera-only detections are deliberately excluded from range-based fusion when there is no valid LiDAR return.

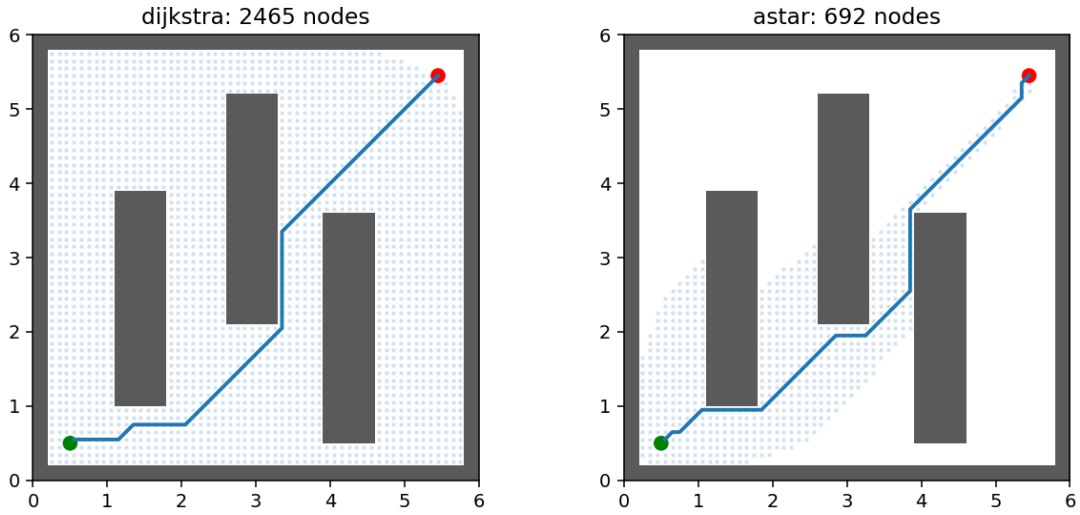


Figure 2. Dijkstra (left) and A* (right) on the same inflated binary grid; blue search cells and their path are computed offline.

Dijkstra: 7.691 m, 2465 expanded nodes, 9.984 ms median. A*: 7.691 m, 692 nodes and 3.516 ms. Both found the same optimal grid cost, while the admissible heuristic reduced expansions by 71.9%. Python timings are device-specific; the relative node count is more portable.

Component A: control and navigation

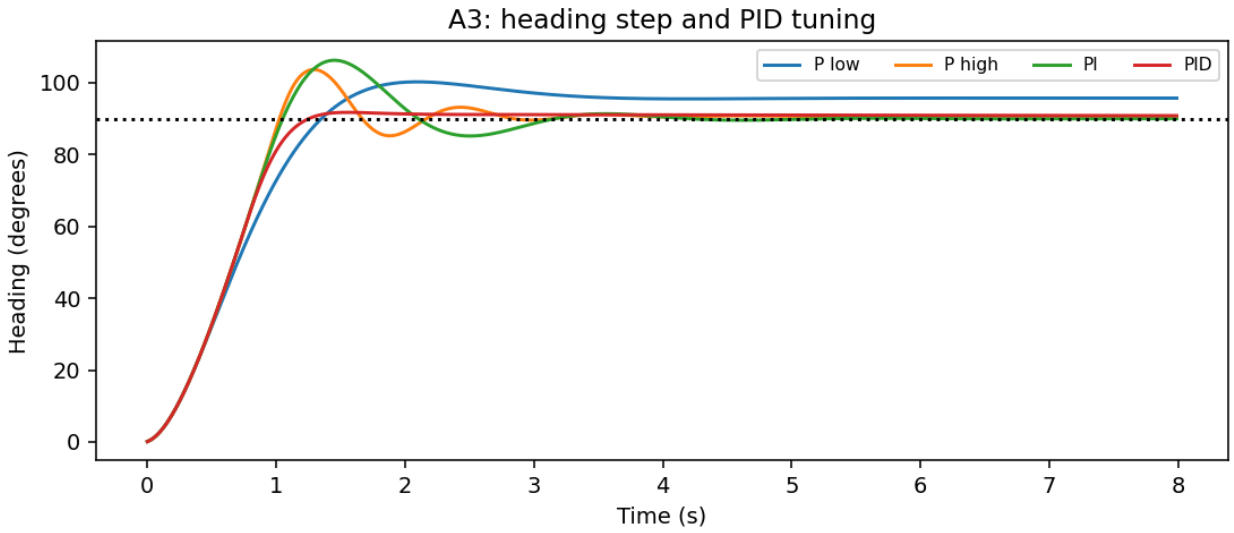


Figure 3. Heading response to a 90° step under an assumed actuator lag and disturbance. Overshoot is assessed against the reference heading.

The proportional-only low-gain model retained 5.729° steady-state error and did not remain within the 2% settling band. PI reduced steady-state error to 0.007°, but overshoot reached 18.07%. The tuned PID (10, 0.6, 2) limited overshoot to 1.95% and settled in 1.17 s. More aggressive control shortens response but can amplify noise or saturate the motors.

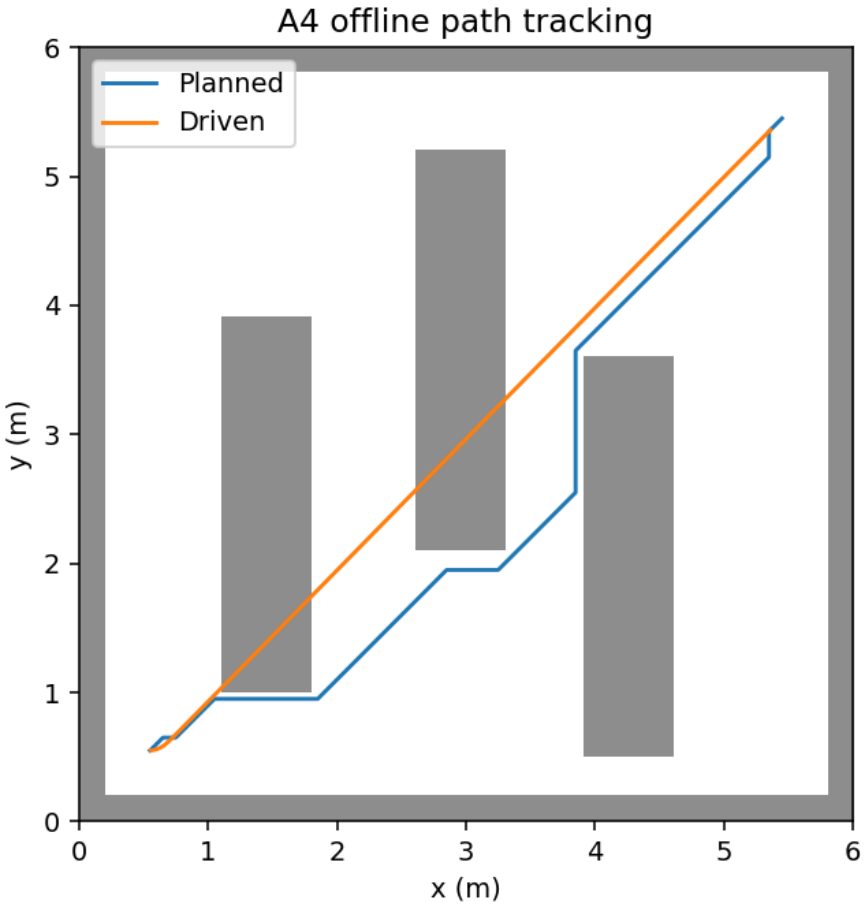


Figure 4. Known-map offline planned and driven trajectories. Replanning, dynamic obstacles and on-robot sensing are implemented separately for Webots but unverified here.

The offline follower reached the defined goal tolerance: True. Initial grid plan = 7.691 m; distance driven = 6.81 m; simulated duration = 45.45 s; RMS discrete reference error = 0.441 m; final error = 0.125 m. The drive can stop before the final cell centre, explaining a shorter actual distance. The relatively large RMS indicates sparse waypoint tracking;

Component B: methods (B1–B3)

B1: Cooperative setting. The supplied Webots world contains four named peer robots in a 3 × 3 m square with obstacle boxes. Each robot has two wheel motors, eight infrared-style proximity sensors, an IMU, GPS, emitter and receiver. The task is area coverage: spread agents across free space rather than converge on the same target. The intended radio network uses range-limited broadcast on channel 1 (1 m), so link availability depends on pairwise distance. Four local controller copies write separate pose, action and estimate logs.

B2: Consensus and swarm search. Offline average consensus runs on a connected four-agent ring. Each agent updates $x_i \leftarrow x_i + \varepsilon\Sigma_j(x_j - x_i)$ for $\varepsilon = 0.45$, below 1/maximum-degree = 0.5. The plotted history quantifies convergence. Particle Swarm Optimisation searches for four station coordinates minimizing the mean squared distance of 240 reference locations to their nearest station. It compares global-best and ring-neighbour best variants (28 particles, 65 iterations, inertia 0.72, cognitive/social gains 1.5), plus a seeded random placement baseline. The ring solution provides nominal target coordinates for the Webots peer controllers. The optimisation is done offline, not collaboratively reoptimised in the Webots controller.

B3: Neural navigation. A two-hidden-layer (16,16) MLP imitates a rule-based expert that maps eight normalised proximity inputs to forward, left turn and right turn. We generated 9,000 synthetic training examples, stratified into an 80/20 split, fixed the random seed and measured held-out accuracy and recall. The model is saved for Webots controller inference; an emergency threshold override replaces unsafe forward commands. These are synthetic proxy inputs rather than readings from the provided Webots world.

Comparative decision-making. A separate seeded 20 × 20-grid proxy contrasts centralised periodic global reassignment with distributed local exploration. It records cells visited and approximate communication messages. The centralised proxy has privileged global knowledge; its stronger performance should not be attributed solely to architecture without accounting for information availability, update period and communication costs. The Webots controller and proxy therefore address different evaluation levels.

Reproducibility controls. Seeds, algorithm definitions, world geometry, plot scripts, report and experiment metrics are bundled with the project. Device orientations, proximity scaling and LiDAR beam order require calibration on the installed Webots release before results are used to support physical deployment claims.

Component B: coordination and learning results

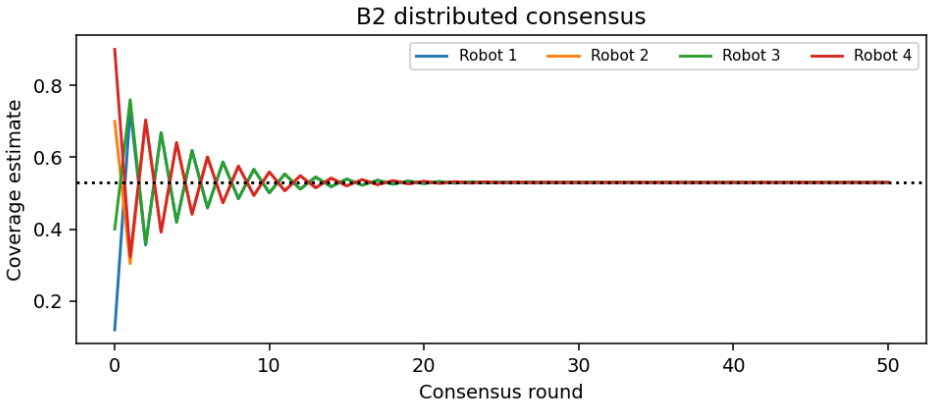


Figure 5. Four ring-connected agents converge toward their initial mean: consensus emerges from repeated neighbour exchange.

Final consensus spread = 7.70714e-06, demonstrating convergence of the connected, static benchmark. The real 1 m radio graph could become disconnected; this proof does not establish convergence under intermittent contact.

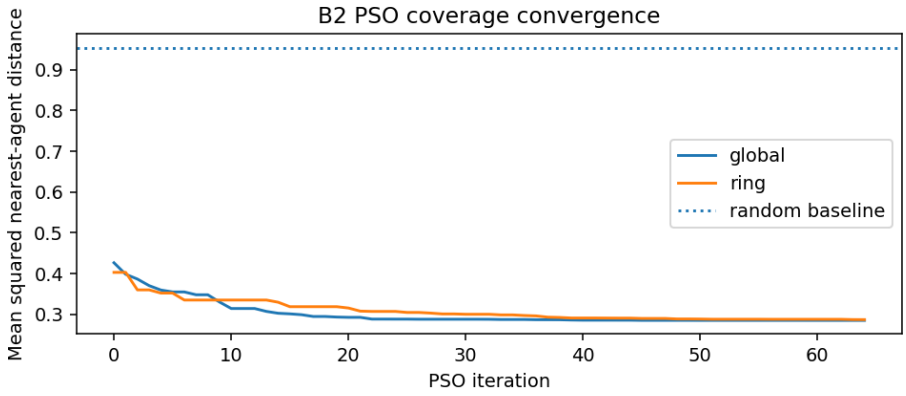


Figure 6. Global- and ring-best PSO reduce squared coverage cost relative to a seeded random baseline.

Final coverage cost: global 0.28547, ring 0.28735 versus random 0.95394. These are mean squared distances, not percentages of area covered; the cost function encourages dispersion but does not model collisions or motion constraints.

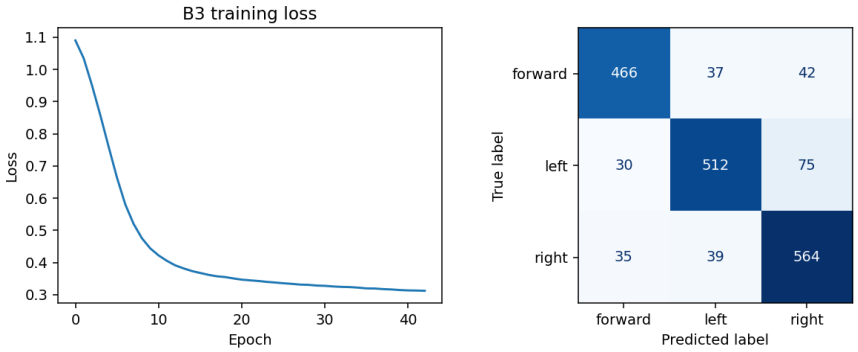


Figure 7. Synthetic training loss and confusion matrix on the held-out set. Minority action recall is reported rather than relying on accuracy alone.

Held-out accuracy = 85.67% (n=1800); class recalls: forward 0.855, left 0.830, right 0.884. Synthetic accuracy is not a robot crash-free success rate. The offline coverage proxy achieved centralised 100.0% vs distributed 89.09% after 160 steps, with mean messages 640.0 vs 131.25 (Figure B3_coordination.png).

B4: ethics, risk and responsible autonomy

1. Bias and reliability. Synthetic camera images use a fixed red colour, the LiDAR noise model is idealised and the MLP is trained on stylised eight-sensor vectors. The observed held-out accuracy cannot establish safe performance with sunlight, reflections, dust, people, wet floors, dark obstacles or locally calibrated sensors. Domain shift should be addressed with varied Webots scenes, genuine sensor logs, held-out arenas and balanced hazards; false negatives near people deserve stronger penalties than harmless detours.

2. Transparency and explainability. A rule-based controller has inspectable thresholds, whereas an MLP can produce a turn from many hidden activations. Preserve each sensor vector, model version, confidence estimate, action and safety override. Compare predictions with the rule-based expert and sample borderline inputs. Human operators should receive plain-language alerts, rather than treating network classification confidence as an explanation.

3. Safety in shared spaces. A wrong turn or missed close object could injure workers or bystanders. Proposed safeguards include speed limits, exclusion areas, hard-wired emergency stop, proximity-based braking, watchdog timeouts, redundant obstruction checks and safety validation beyond a simulation. The supplied emergency threshold is a software illustration and does not constitute a certified safety function.

4. Fault tolerance. The dynamic 1 m communication topology can partition. When packets disappear, an agent should retain a time-stamped local map, abandon stale consensus values and avoid unverified destinations. A failed robot should be removed from allocation, while survivors redistribute coverage. Test message loss, dead batteries, sensor dropout and processor stalls with repeated seeded runs and record post-failure coverage and minimum separation. The submitted offline ring-consensus convergence result assumes connectivity; it does not cover these faults.

5. South African context. Where identifiable video or location traces are captured, Protection of Personal Information Act (POPIA) duties require lawful purpose, information minimisation, access controls and retention decisions. Industrial deployment also requires appropriate workplace safety assessment under the Occupational Health and Safety Act. Local deployments must account for power instability, uneven networks, dust in mining or farming, language-inclusive warnings and workforce retraining. Human oversight, meaningful incident reporting and stakeholder consultation are more defensible than replacing responsibility with the phrase “the AI decided”.

Ethical position. The present experiment is suitable for demonstrating algorithms in simulation only. Deployment needs separately documented hazard analysis, incident escalation, independent verification and measurable safety acceptance criteria. This is consistent with the module guide’s treatment of multi-agent robustness and the ethical implications of autonomy (Richfield Graduate Institute of Technology, n.d.b, Units 5–6).

Conclusion, recommendations and references

Conclusion. Component A demonstrates offline LiDAR/camera evidence fusion, admissible A* versus Dijkstra, tunable PID response and known-map closed-loop path following. A* reduced expansions substantially without sacrificing this map's optimal path length. The PID suppressed overshoot relative to PI, but the measured tracking error and shortcutting of waypoints indicate the need to tune tracking and safety margins. Component B demonstrates four-agent consensus, PSO-based dispersion and neural imitation of a proximity-navigation expert. The offline proxy highlights a centralised coordination advantage bought with global information and greater message count.

Recommendations and limits. Before submission, execute both .wbt worlds in Webots; inspect wheel direction, sensor names, 360° beam order, camera bearing sign and contact geometry; run a 120-s swarm trial; save two world screenshots, logs and failure tests; retrain B3 on recorded Webots samples and measure closed-loop collision-free success. Repeat multiple seeded trials, report actual Webots version and hardware, and update the results tables with simulator-derived evidence. These actions have not been carried out in the delivered files. The accompanying README explicitly separates verified offline results from unverified simulator integration.

References (Harvard style)

Richfield Graduate Institute of Technology (n.d.a) Autonomous Systems and Robotics 700: Assignment instructions — Intelligent Autonomous Robotic Systems Project. Course assignment brief, pp. 1–7.

Richfield Graduate Institute of Technology (n.d.b) Autonomous Systems and Robotics 700: Learner Guide. Faculty of Information Technology, Units 2–6, pp. 23–353.

Richfield Graduate Institute of Technology (n.d.c) ASR700 Assignment Blueprint. Course slide deck, slides 1–51.

Republic of South Africa (1993) Occupational Health and Safety Act 85 of 1993. Pretoria: Government Printer.

Republic of South Africa (2013) Protection of Personal Information Act 4 of 2013. Pretoria: Government Printer.

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